

Process name:	Vehicle Contract Automation	Platform:	n8n
Type:	Learning Project / PoC	Domain:	AI Data Extraction
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Problem	The billing team at a car dealership used to open each PDF contract and manually type the vehicle data into a spreadsheet — a slow process prone to typing errors.
Description	Workflow that reads vehicle purchase contracts in PDF, extracts 8 fields using an AI model, and automatically logs them into a spreadsheet.
Purpose	Eliminate manual transcription of contract data, reducing time and typing errors.
Architecture	Google Drive source folder (PDFs) → n8n orchestration, 10 nodes → Google Gemini Information Extractor → Google Sheets append-or-update → Gmail final notification

Flow	#	NODE	TYPE	WHAT IT DOES	RECEIVES → RETURNS	WHY IT'S NEEDED
	1	When clicking "Execute workflow"	Manual start	It's the button that starts the process.	— → the process starts running	In this test it's started manually; in real use it would trigger on its own, for example every day at a set time or as soon as a new contract arrives in the folder.
	2	Search files and folders	Google Drive	Searches and lists all the PDFs stored in the contracts folder.	Folder → 15 files found	It tells the process which contracts it needs to process in that run.
	3	Loop Over Items	Loop	Takes the 15 contracts and processes them one by one, repeating the following steps for each.	15 files → 1 contract at a time	If one contract fails, it doesn't affect the others. Also, once all of them are processed, an email is sent notifying that the process is complete.
	4	Download file	Google Drive	Downloads the PDF contract it's currently processing.	File name → downloaded PDF file	To read the contract's content, it first needs to be downloaded.
	5	Extract from File	Extract From PDF	Converts the PDF content into readable text.	PDF file → contract text	The AI model that extracts the data needs text, it can't read a PDF directly.
	6	Code in JavaScript	Text cleanup	Cleans and organizes the contract text before sending it to the AI.	Unsorted text → clean text	It improves the accuracy with which the AI identifies the vehicle data.
	7	Wait	Wait	Adds a short pause before querying the AI with the next contract.	— → a brief pause	The AI only allows a limited number of requests per minute; this pause prevents the process from failing by exceeding that limit.
	8	Information Extractor	AI (Gemini)	Reads the contract text and extracts the 8 vehicle data fields.	Contract text → the 8 vehicle fields, already organized	The AI interprets the text extracted from the document to extract the vehicle data.
	9	Append or update row in sheet	Google Sheets	Saves the contract data as a new row in the spreadsheet; if that vehicle was already registered, it updates its row instead of creating a duplicate.	The 8 fields → one row in the sheet	It prevents duplicate contracts if the process is run more than once.
	10	Send a message	Gmail	Sends an email letting the team know all contracts have finished processing.	End of process → email sent	This way the billing team doesn't need to check the sheet to know whether it's done.
Technical decision: AI model	Function: Gemini is the model used by the Information Extractor node (step 8). Based on the contract text, it identifies and returns the 8 required fields in structured format, replacing manual transcription. Selection criteria: • Cost: the free tier is enough for this workload, with no need to pay for a plan. • Context window: allows processing multi-page contracts in a single call, with no need to split them.					
Results and potential impact	The automation eliminates manual transcription of contract data, reducing the risk of typing errors and freeing up the billing team's time for higher-value tasks.					